

NAKIB UDDIN AHMED

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#) | [Google Scholar](#)

RESEARCH INTERESTS

CNN-Transformer Hybrid Architectures · Explainable & Trustworthy AI · Medical Image Analysis · Uncertainty Quantification · Multimodal Spatial AI · Transfer Learning & Efficient Deep Learning · Computer Vision · Machine Learning · Healthcare AI · Ensemble Learning · Multimodal Learning & Ensemble Methods · Geospatial AI & Urban Analytics

RESEARCH PROFILE

AI Researcher and Machine Learning Engineer (M.Sc. Information Technology, CGPA 3.88/4.00) with **2 published**, **1 conference-accepted**, and **3 under-review** peer-reviewed papers in Springer Nature, Elsevier, and IJCA. Specializing in CNN-Transformer hybrid architectures, medical image analysis, and explainable AI, with applications spanning gastrointestinal endoscopy risk stratification, polyp segmentation, brain tumor classification, maternal and fetal health, housing affordability forecasting, and online recruitment fraud detection.

RESEARCH STATEMENT

My research develops explainable CNN-Transformer hybrid architectures for high-stakes real-world decision support, spanning clinical medical imaging and urban policy forecasting, where model transparency and reliability are as critical as predictive accuracy. In the medical domain, I design lightweight CNN-Transformer encoders with uncertainty quantification (Monte Carlo Dropout) and visual explainability (Grad-CAM, attention maps) to support trustworthy clinical decision-making in gastrointestinal endoscopy and polyp segmentation. In the urban domain, I extend this interpretability-first approach to multimodal spatial AI, combining street-level and satellite imagery with socioeconomic tabular features in a CNN-Transformer framework to produce explainable, SHAP-driven affordability forecasts that inform evidence-based housing policy. Going forward, I aim to unify these two threads by building generalizable, uncertainty-aware explainable AI systems that bridge visual perception and structured data reasoning, deployable across both clinical environments and societal decision-making contexts where the cost of an opaque prediction is unacceptably high.

EDUCATION

Master of Science in Information Technology

Shahjalal University of Science and Technology (SUST), Bangladesh

2023 – 2025

CGPA: 3.88/4.00

Relevant coursework: Machine Learning, Artificial Intelligence, Model Development, Python Algorithms, Data Science, Deep Learning, Database Systems, Information Systems.

Industry-Sponsored Internship & Final Year Project: Developed [Stock Trend Prediction Using Bangla News Analysis](#) during an M.Sc. industry internship at **DataSoft Systems Bangladesh Limited**, one of Bangladesh's leading technology companies.

Bachelor of Science in Computer Science and Engineering

Metropolitan University, Bangladesh

2018 – 2022

CGPA: 3.69/4.00

Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, Machine Learning, Artificial Intelligence, Bioinformatics Computing, Discrete Mathematics.

Final Year Project: [AI-Based E-Commerce Platform](#) — built a production-grade online store using Django and React with a machine learning regression model for real-time laptop price prediction.

HONOURS & AWARDS

- **First Position — AI Innovation Challenge** — Idea Sharing 2020, organized by Hours of Code at Metropolitan University.
- **5+ Peer-Reviewed Publications** — Published and accepted research in Springer Nature, Frontiers, Elsevier, and IJCA spanning medical imaging, computer vision, NLP, and machine learning.
- **Multidisciplinary Certifications** — Certified across data analytics, ML, Python, SQL, and BI from Interactive Cares, HackerRank, Kaggle, and Google Cloud.

PUBLICATIONS

AI-Driven Housing Affordability Forecasting in New York City: An NTA-Level Panel Analysis Using Ensemble Machine Learning

Nakib Uddin Ahmed, Azizur Rahman, Mehjabin Ferdous

Applied Spatial Analysis and Policy, Springer Nature, 2026

[Published] | [Paper Link](#)

- Built an explainable AI framework predicting severe housing cost burden across NYC neighbourhoods using socio-economic, housing, and rental market data (2012–2022).
- LightGBM achieved $R^2 = 0.93$ and $RMSE = 0.0337$ under strict temporal cross-validation; applied SHAP explainability for policy-level feature attribution.

Beyond Binary: Four-Class Risk Stratification from Gastrointestinal Endoscopy Using Asymmetric-Cost Lightweight CNN–Transformer Learning

Nakib Uddin Ahmed, Azizur Rahman, Mehjabin Ferdous

Evolutionary Intelligence, Springer, 2026

[Under Review] | [Paper Link](#)

- Designed a lightweight CNN–Transformer framework for four-class gastrointestinal lesion risk stratification aligned with ACG and ESGE clinical guidelines.
- Evaluated DenseNet-121, EfficientNet-B0, and DeiT-Tiny with a novel **Asymmetric Endoscopy Loss (AEL)** and MC-Dropout uncertainty estimation; achieved **macro F1 = 0.84** with **zero missed high-risk lesions**, automating triage of **44.9% low-risk cases**.

Enhancing Online Recruitment Fraud Detection: A Comparative Analysis of Gradient Boosting and Transformer Architectures Under Severe Class Imbalance

Nakib Uddin Ahmed, Azizur Rahman

International Journal of Computer Applications, 2024

[Published] | [Paper Link](#)

- Developed an AI-based fraudulent job-posting detection system using Machine Learning, Deep Learning, and Transformer models on **17,883 job advertisements** with Word2Vec embeddings and SMOTE.
- XGBoost reached **99.44% accuracy** and **F1 = 0.99**, demonstrating the effectiveness of ensemble learning and transformer-based approaches for fraud detection.

PolyDetect: A Quality-Aware Hybrid Deep Learning Framework for Robust Polyp Segmentation in Degraded Colonoscopy Images

Nakib Uddin Ahmed, Azizur Rahman

International Journal of Medical Informatics, Elsevier, 2026

[Under Review] | [Paper Link](#)

- Proposed PolyDetect integrating image quality assessment, adaptive image enhancement, and a hybrid Transformer–CNN architecture for robust colorectal polyp segmentation under image degradation.
- Achieved **Dice = 0.8066** and **IoU = 0.7262** while running at **53.62 FPS**, enabling accurate and clinically applicable real-time polyp detection.

Soft Voting Ensemble Learning and SHAP-Based Interpretability for Maternal and Fetal Health Risk Stratification Across Multiple Clinical Datasets

Nakib Uddin Ahmed

IEEE Xplore, 2026

[Accepted] | [Paper Link](#)

- Developed a soft-voting ensemble framework combining XGBoost and Random Forest for maternal and fetal health risk stratification across multiple clinical datasets.
- Applied **SHAP-based interpretability** with 5-fold stratified cross-validation to identify key clinical drivers of high-risk pregnancy, supporting explainable clinical decision-making.

Cross-Domain Transfer Learning for Brain Tumor Classification Under Limited MRI Data Regimes

Nakib Uddin Ahmed, Azizur Rahman, Mehjabin Ferdous

Medical & Biological Engineering & Computing, Springer, 2026

[Under Review] | [Paper Link](#)

- Evaluated transfer learning for four-class brain tumor MRI classification using EfficientNet-B0 across varying training data sizes, comparing ImageNet-pretrained and randomly initialized models on **7,200 MRI images**.
- Demonstrated significant gains under extreme data scarcity with **Grad-CAM++ explainability**, providing practical insights for data-efficient AI in medical imaging.

PROFESSIONAL EXPERIENCE

Executive, Data Analyst | Bionic, Bangladesh

Jan 2025 – Present

- Analyze sales and operational data to define KPIs and deliver actionable insights for strategic decision-making.

- Built **revenue forecasting models** using predictive machine-learning techniques to support growth planning.
- Designed and maintained automated **BI dashboards** to monitor performance and improve reporting efficiency.
- Developed merchant segmentation and anomaly detection models to enhance engagement and operational effectiveness.

Web Developer and CRM Administrator | Global Education Care

Jul 2023 – Dec 2024

- Designed, built, and optimized responsive websites using modern web technologies.
- Managed, customized, and maintained CRM systems to enhance user experience and efficiency.
- Ensured data integrity, automated workflows, and generated reports for business insights.
- Provided support, troubleshooted issues, and trained users on CRM and web applications.

Machine Learning Engineer (Internship) | DataSoft Systems Bangladesh Limited

Jun 2023 – Nov 2024

- Undertaken as part of the **M.Sc. university–industry collaboration** between SUST and DataSoft Systems Bangladesh Limited, one of Bangladesh’s leading technology companies.
- Developed a **Stock Trend Prediction model** integrating news sentiment analysis with financial data for trend forecasting.
- Implemented NLP techniques (word embedding, sentiment classification) to analyze financial news impact on stock movements.
- Utilized Scikit-learn and ensemble learning models to enhance prediction accuracy with visualized insights.

IT Instructor and IT Support Technician | UCEP BANGLADESH

Jun 2022 – Apr 2023

- Diagnosed and resolved hardware, software, and network issues.
- Instructed and mentored school and diploma students in IT-related subjects.
- Conducted training sessions on various IT skills and equipped students with ICT capabilities.

SELECTED TECHNICAL PROJECTS

MedicalVision-AI: Medical Imaging with XAI | [GitHub](#) | *PyTorch, FastAPI, EfficientNet-B0, Grad-CAM*

- Built a unified platform consolidating **16 CNN-based medical imaging classification projects** with **7 explainable AI methods** (Grad-CAM, Grad-CAM++, Score-CAM, Integrated Gradients, and more) served through a FastAPI dashboard with downloadable PDF reports.

Breast Cancer AI Classifier: Advanced FNA Analysis | [GitHub](#) | *XGBoost, Random Forest, Scikit-Learn, FastAPI*

- Trained multiple classifiers on the Wisconsin Breast Cancer dataset (30 FNA features) with threshold tuning to achieve **98%+ recall** on malignant cases, minimizing false negatives in a screening context.

Stock Trend Prediction Using Bangla News Analysis | [GitHub](#) | *Bangla NLP, Random Forest, FastAPI*

- Predicted stock market trends via sentiment analysis of Bangla financial news, classifying articles as positive, negative, or neutral; achieved **98.43% accuracy** with Random Forest.

Arcane Chess: Deep Reinforcement Learning Chess | [GitHub](#) | *PyTorch, Deep Q-Network, FastAPI*

- Implemented a deep reinforcement learning chess platform with a **DQN agent** featuring self-attention, dueling architecture, and prioritized experience replay.

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript, TypeScript, Java, C++, C
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, TensorFlow, Pandas, NumPy, SHAP
- **Deep Learning:** PyTorch, CNNs, Vision Transformers, Transfer Learning, Multimodal Learning, MambaOut
- **Research Methods:** Computer Vision, NLP, Explainable AI, Imbalanced Learning, Feature Engineering, Cross-Validation, Temporal Validation, Spatial Analysis
- **Medical AI:** Medical Imaging, Clinical Decision Support, Risk Stratification, Uncertainty Quantification
- **Web & Mobile:** Django, FastAPI, React.js, Next.js, React Native, Node.js
- **Databases:** PostgreSQL, MySQL, MongoDB, Firebase
- **Tools:** Git, GitHub, Docker, Jupyter Notebook, REST APIs

LANGUAGES

- **English:** Fluent
- **Bengali:** Native
- **Hindi:** Conversational

REFERENCES

M. Jahirul Islam, PhD., PEng.

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